

BARUUNTURUUN, Mongolia

WMO: 442130

Lat: 49.6586N

Lon: 94.4042E

Elev: 4042

StdP: 12.67

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.5	-37.8	-47.5	0.4	-41.3	-43.3	0.5	-37.4	10.6	-10.7	9.2	-16.1	0.1	120	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	86.3	62.0	82.8	60.5	79.6	59.3	66.2	80.1	63.7	77.8	61.6	75.2	6.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.0	93.1	74.6	58.3	84.6	71.4	55.9	77.4	68.9	33.4	80.4	31.3	77.6	29.6	75.4	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	19.0	15.7	12.5	DB	-41.5	93.2	5.0	5.0	-45.1	96.7	-48.0	99.6	-50.8	102.4	-54.5	106.0
(5)				WB	-41.0	68.2	5.4	4.5	-44.9	71.5	-48.0	74.1	-51.0	76.7	-54.9	79.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	-21.9	-16.2	4.0	36.0	51.3	62.6	66.6	62.4	50.3	34.4	11.1	-11.3
	DBStd	32.69	7.31	9.97	11.51	12.07	9.14	7.35	6.15	6.88	8.57	9.63	10.40	9.16
	HDD50	9698	2227	1853	1425	436	96	6	0	3	99	486	1166	1901
	HDD65	13846	2692	2273	1890	871	433	128	53	131	446	947	1616	2366
	CDD50	1544	0	0	0	15	135	383	513	386	108	4	0	0
	CDD65	215	0	0	0	0	6	55	101	49	4	0	0	0
	CDH74	2386	0	0	0	0	97	664	1038	544	43	0	0	0
	CDH80	650	0	0	0	0	12	165	320	148	5	0	0	0
Wind	WSAvg	3.3	0.8	1.0	2.0	5.0	6.4	5.3	4.5	4.0	4.3	3.6	1.9	1.1
Precipitation	PrecAvg	8.6	0.2	0.2	0.3	0.6	0.7	1.2	2.0	1.6	0.9	0.5	0.4	0.3
	PrecMax	12.3	0.5	0.5	0.7	1.4	2.1	3.3	5.7	4.6	2.5	1.5	1.2	0.6
	PrecMin	4.3	0.0	0.0	0.0	0.1	0.0	0.1	0.2	0.2	0.2	0.0	0.0	0.0
	PrecStd	1.7	0.1	0.1	0.2	0.4	0.5	0.7	1.3	0.9	0.6	0.4	0.3	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.6	16.5	41.0	69.1	81.8	88.9	92.7	89.9	78.5	63.2	41.3	17.7
		MCWB	-0.5	14.4	33.4	48.7	56.6	61.2	67.8	63.5	56.5	46.3	34.7	15.2
	2%	DB	-3.9	9.0	31.6	64.0	76.5	84.5	87.3	83.9	74.0	58.5	34.8	10.1
		MCWB	-4.8	7.6	27.9	47.7	54.8	59.6	63.8	61.2	54.9	44.2	29.9	8.8
	5%	DB	-7.3	2.6	26.2	59.1	72.3	81.0	83.6	79.9	70.0	54.1	30.6	5.8
		MCWB	-7.9	1.6	23.6	44.9	52.9	59.0	61.7	59.9	52.8	42.0	27.4	4.9
	10%	DB	-10.4	-1.7	21.4	54.5	68.2	77.7	79.8	76.1	65.7	49.7	26.3	2.1
		MCWB	-10.9	-2.6	19.4	42.6	50.8	57.9	60.3	58.7	50.7	39.6	23.9	1.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.3	14.3	34.1	52.2	60.7	65.6	70.9	67.6	60.0	48.3	35.2	15.6
		MCDB	0.5	16.3	41.5	65.3	76.1	80.5	88.6	85.5	76.5	60.3	40.6	17.1
	2%	WB	-4.7	7.5	28.0	48.1	57.4	63.4	67.6	64.1	56.4	45.4	30.5	9.0
		MCDB	-4.0	9.0	31.4	62.2	72.6	79.4	82.6	79.4	70.6	56.4	33.8	10.1
	5%	WB	-7.9	1.7	23.7	45.5	54.4	61.4	65.2	61.9	53.7	42.6	27.5	4.8
		MCDB	-7.3	2.7	26.0	58.2	69.7	77.0	78.3	76.2	67.6	53.4	30.5	5.6
	10%	WB	-10.9	-2.5	19.6	42.9	51.8	59.3	62.8	59.9	51.6	40.1	23.8	1.3
		MCDB	-10.5	-1.7	21.4	53.7	66.5	74.2	75.2	72.8	63.9	49.2	26.3	2.0
Mean Daily Temperature Range	5% DB	MDBR	16.7	19.0	20.7	18.7	21.7	21.6	19.8	19.8	21.0	19.4	16.4	15.3
		MCDBR	18.0	21.5	20.3	24.0	26.0	26.3	25.8	25.4	25.1	24.7	17.5	17.1
		MCWBR	17.4	20.2	17.0	13.2	13.0	11.9	11.5	11.9	12.6	15.1	13.6	16.1
	5% WB	MCDBR	17.9	21.6	19.8	22.8	24.7	23.2	22.5	22.5	23.5	23.7	16.8	17.0
		MCWBR	17.3	20.3	16.6	14.0	14.4	13.0	12.5	12.4	13.0	15.2	13.5	16.0
Clear-Sky Solar Irradiance	taub	0.242	0.268	0.312	0.353	0.358	0.358	0.358	0.339	0.304	0.288	0.267	0.244	
	taud	2.284	2.230	2.208	2.248	2.268	2.359	2.389	2.415	2.462	2.428	2.325	2.256	
	Ebn at Noon	261	279	283	282	285	285	284	285	285	270	247	241	
	Edh at Noon	22	30	36	40	41	37	36	33	28	24	21	19	
All-Sky Solar Radiation	RadAvg	451	782	1286	1725	2080	2167	2016	1754	1410	889	469	344	
	RadStd	34	40	42	87	97	79	111	87	76	54	39	33	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	-3.91	-5.99	N/A	-2.09	-2.47	N/A	N/A	N/A	N/A
Regional (5 neighbors)		N/A	-3.51	-4.82	N/A	-1.55	-1.62	N/A	N/A	N/A	N/A

Nomenclature: See separate page